UX/UI & System Architecture Overview

Auroville Eco-Service ERP — User Interface, Workflows & Operational Architecture

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Introduction: ERP Architecture, Workflows & Atomic Ledger

Conceptual groundwork behind data storage, process execution, and long-term operational autonomy.

- **ERP & Systemic Workflow Architecture:** Enterprise Resource Planning (ERP) in this context serves as a single source of truth for physical material flows. Data storage is organized around an immutable event-driven workflow where waste bags transit through discrete operational stages (submission, pickup, weighing, sorting, and fraction allocation) without manual tallying.
- **Atomic Business Operations:** Every physical action is recorded as an indivisible, immutable transaction (an atomic event, e.g., `BagWeighed`). By preserving exact timestamps, weights, operators, and spatial tags, the system prevents data corruption, masks individual errors, and maintains flawless accounting integrity across all processing hubs.
- **Operational Liberation through Granular Accounting:** Detailed granular logging initially captures existing field realities. Over time, as historical data accumulates, algorithmically revealed patterns automate routing, baseline predictions, and anomaly detection—gradually relieving operators, managers, and residents from routine cognitive load and administrative friction.

1. System Entry & Mobile Portals

Main role selection desktop portal alongside mobile-optimized apps for field collectors and hub sorters.

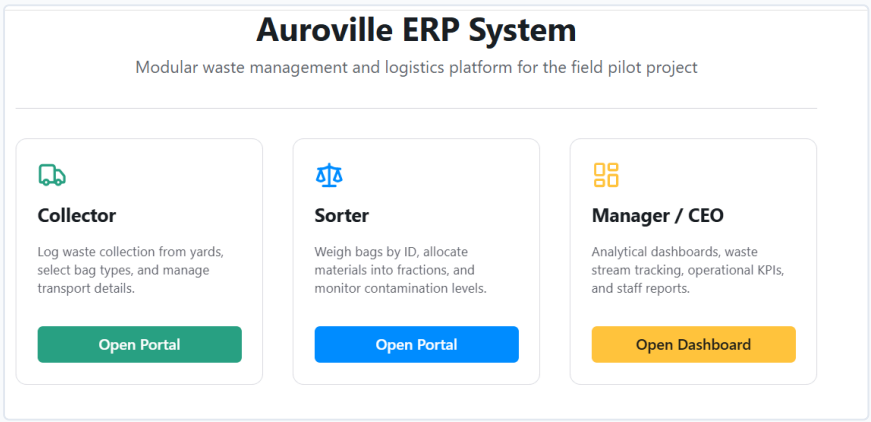


Figure 1.1: Main Screen — Role Navigation Portal

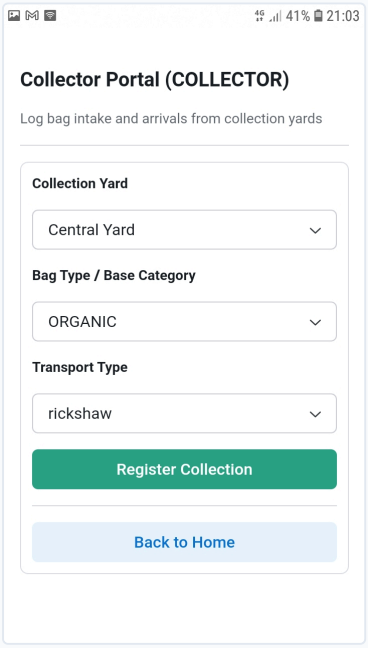


Figure 1.2: Field Collector Interface

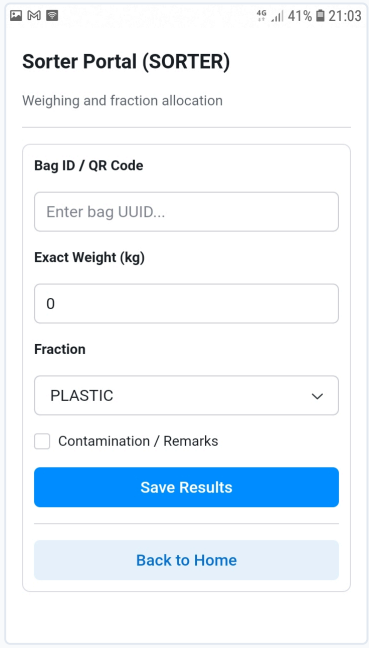


Figure 1.3: Hub Sorter Interface

2. Management Dashboard Overview & Operator Activity

Real-time system overview, daily mass trends, and operator performance logs.

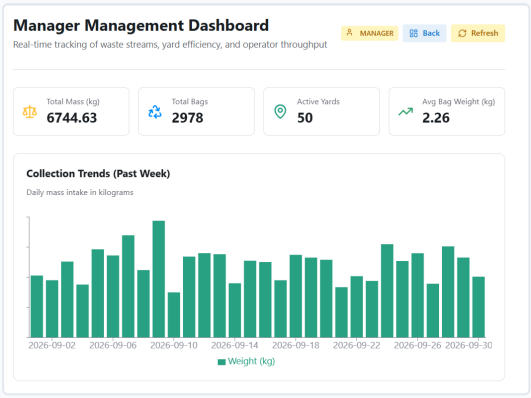


Figure 2.1: Executive Dashboard & Collection Trends

Operator Performance

Field staff operations activity logs and throughput volume

Operator	Role	Event Type	Operations
Vijay	COLLECTOR	BagArrivedAtHub	1049
Raman	COLLECTOR	BagArrivedAtHub	970
Anand	COLLECTOR	BagArrivedAtHub	959
Saraswati	SORTER	BagWeighed	775
Saraswati	SORTER	BagSorted	775
Devi	SORTER	BagSorted	755
Devi	SORTER	BagWeighed	755
Selvi	SORTER	BagWeighed	730
Selvi	SORTER	BagSorted	730
Lakshmi	SORTER	BagSorted	718
Lakshmi	SORTER	BagWeighed	718

Figure 2.2: Operator Performance Logs & Activity

3. Waste Fractions & Yard Performance Breakdown

Material structure breakdown by fraction percentages and community yard volume throughput.

Waste Structure by Fractions

Breakdown by categories, codes, weight, bags, and percentages

– Scroll horizontally to view all data

Code	Material Fraction	Bags	Total Weight (kg)	Share (%)
ORGANIC	Food waste, solid (Organic)	1273	2657.123	39.4 %
PAPER	Paper waste (Recyclable)	280	1564.593	23.2 %
CARDBOARD	Cardboard (Recyclable)	198	880.524	13.06 %
PLASTIC	Plastic (Recyclable)	373	577.837	8.57 %
SANITARY	Sanitary waste (Non-recyclable)	338	352.514	5.23 %
GLASS_UNBROKEN	Glass, Unbroken (Recyclable)	116	262.126	3.89 %
NON_RECYCLABLE_KITCHEN	Other waste (kitchen) (Non-recyclable)	299	258.784	3.84 %
METAL	Metal (Aluminium, Copper, etc.) (Recyclable)	35	82.92	1.23 %
TYRES	Tyres (Special Processing)	2	52.998	0.79 %
TEXTILE	Used textile (Non-recyclable)	5	28.928	0.43 %
TETRAPACK	Tetrapack (Recyclable)	32	9.991	0.15 %
GLASS_BROKEN	Glass, Broken (Recyclable)	16	9.081	0.13 %
THERMOCOL	Thermocol / Styrofoam (Recyclable/Processed)	9	6.53	0.1 %
BATTERIES	Batteries (Hazardous/E-Waste)	2	0.681	0.01 %

Figure 3.1: Material Structure

Community Yard Performance

Total mass collected and bag counts by collection points

Community / Yard	Address	Bags Received	Total Weight (kg)	Avg Bag Weight (kg)
Fraternity	Fraternity Community, Auroville	51	206.905	4.06
Discovery	Discovery Community, Auroville	71	195.403	2.75
Anuyog	Anuyog Community, Auroville	69	186.246	2.7
Alenfels	Alenfels Community, Auroville	70	183.53	2.62
City Light	City Light Colony, Auroville	65	177.89	2.74
Awaken	Awaken Community, Auroville	68	176.657	2.6
Heritage	Heritage Village, Auroville	61	164.282	2.69
Ami Community	Ami, Auroville	69	163.887	2.38
Bottle Centre Point	Bottle Centre, Kuilapalayam, Auroville	57	158.879	2.79
Transformation	Transformation Area, Auroville	50	158.682	3.17
La Ferme	La Ferme Community, Auroville	62	155.163	2.5
Discipline	Discipline Community, Auroville	56	153.447	2.74
Druk	Druk Community, Auroville	59	153.437	2.6
Chinmaya	Chinmaya, Auroville	54	153.301	2.84
Aureole	Aureole, Auroville	62	152.136	2.45
Vikas Community	Vikas, Auroville	70	150.629	2.15
Green Zen	Green Zen Community, Auroville	63	148.736	2.36
Joy Community	Joy, Auroville	62	147.024	2.37
Easy Community	Easy, Auroville	68	145.417	2.14

Figure 3.2: Community Yard Efficiency

4. Mobile Executive Dashboards

On-the-go monitoring of core KPIs and collection intake trends via mobile web.

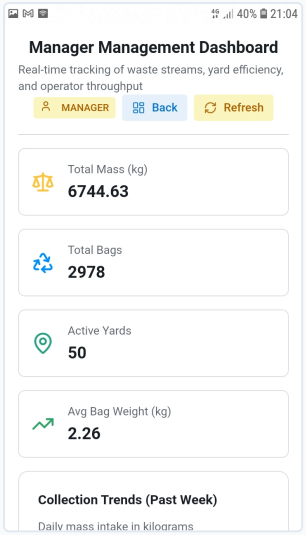


Figure 4.1: Mobile Executive KPIs

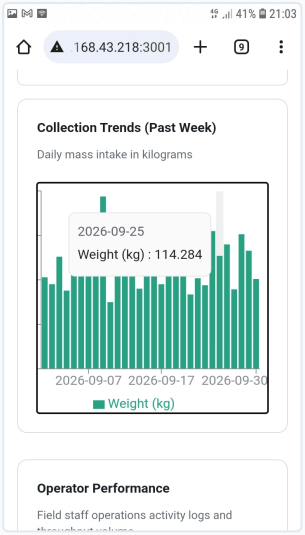


Figure 4.2: Mobile Intake Trends

5. Prototype Context & UI/UX Vision Roadmap

Connecting visual interfaces to underlying business process maturity, system integrity, and long-term automation goals.

- **Test Environment & Demonstration Purpose:** This gallery represents an architectural testbed and conceptual illustration rather than an active operational MVP. Full Minimum Viable Product status will be achieved once the platform is deployed directly into physical operational workflows.
- **UI/UX & Business Process Alignment:**
  - **User Interface (UI):** Focused on radical simplicity and minimal friction. Field screens avoid redundant entry fields, ensuring fast, error-free input in rugged field conditions.
  - **User Experience (UX):** Designed to unburden residents, field collectors, and facility managers from administrative routine, turning waste tracking into an effortless operational step.
  - **Business Process Foundations:** A clean interface directly mirrors management clarity. Well-defined operational workflows form the necessary foundation for structuring data inside **PostgreSQL**, which subsequently dictates clean execution code.
- **Core Architectural & Operational Principles:**
  - **Constraints as Systemic Catalysts:** Technical and physical field limits (lack of connectivity, power outages, limited staff) are treated not as impediments, but as catalysts for deep process ergonomics, offline caching, asynchronous queuing, and a robust backup culture.
  - **Data Anomaly Localization:** Tracking continuous streams over time establishes an objective baseline norm. Any deviations naturally isolate into discrete anomaly zones, enabling precise incident discovery and transparent administrative decisions.
  - **Progressive Adoption & Resident Inclusion:** Process mapping and data logging precede analytics. Initial resident errors are treated as system baselines rather than deviations, building a welcoming interactive UI that users naturally adopt and cultivate over time.
- **Atomic Ledger & Interface Economics:** Pure, immutable event logging allows hidden operational links to surface over time. As these discovery loops strengthen the business core, frontend interfaces become commoditized—their implementation cost and effort approach zero while system value compounds.

6. Technical Architecture & Scaling Philosophy

Database security, append-only event ledgers, dynamic analytical views, and predictive pilot scaling.

- **Schema Segregation & RBAC Security:** Enterprise security in PostgreSQL is not bureaucratic overhead, but the structural core of data integrity. Splitting the database into isolated schemas (`dicts`, `infrastructure`, `security`) and enforcing 4-level Role-Based Access Control guarantees that field collectors cannot mutate weight records, residents only see their own passports, and administrative write vectors remain strictly guarded against asynchronous race conditions.
- **Append-Only Event Store & Dynamic Read Views:** The ledger (`infrastructure.bag_events`) is strictly append-only. An erroneous entry is an immutable historical fact; it is compensated by appending a corrective event rather than rewriting past rows. Because data storage and business reporting have opposing technical requirements, analytical dashboards query dynamic PostgreSQL views that compute material totals on the fly without altering transactional history.
- **Pilot Data & Predictive Simulation (10 to 50+ Sites):** Conducting a 60-day pilot across 10 diverse commercial and residential entities captures representative baseline profiles for the whole settlement. Feeding this real-world audit data into a deductive PL/pgSQL simulation engine allows managers to forecast logistics bottlenecks, vehicle requirements, operator workloads, and precise CAPEX/OPEX metrics before deploying to 50+ sites across Auroville.